HOMEWORK

Review on Chapter 1

- Classification of Ligands:

✓ Based on σ-bonding:

lone pair donor, σ-bong electron pair donor, σ-bonding electron pair donor

✓ Based on π-bonding:

π donor, π acceptor

✓ Most OM ligands are considered as π acceptors

- Notation for hapticity of ligands (η^n)
- Bonding involving lone pair donors, π-bond electron pair donors, and σ-bond electron pair
- Hard-soft acid/base

✓ Hard acids like hard bases, soft acids like soft bases.

✓ Most OM ligands are considered as soft bases.

- Other classifications of ligands

✓ Based on charge: L and X types of ligands, e.g. NH_3 vs Cl^-.

✓ Based on locations: Terminal and bridging ligands

- Valence electrons (VE) count

✓ *Covalent model* : Both M and L are considered as neutral

✓ *Ionic model :* Metal complexes are formed M^{n+} + L + X^-

HOMEWORK

- 18-electron rule and its limitation.
  -***Not all OM complexes follow the rule.***
- Oxidation number
  -***Oxidation number usually can not be higher than group member .***
- Coordination number and geometry
  -**Each C.N. is associated with one or more geometries**
- Effect of complexation
  - **Change of electron density of L**
  - **Change of electron density distribution**
- Differences between metals
  -**Trend in the relative energy of d orbitals**
- Trans effect and trans influence
  - **Highest trans-effect ligands form unusually strong σ/π bonds**

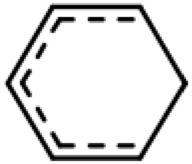
HOMEWORK

1. Based the 18-electron rule, determine the charge, z, of the following complexes.

(i) $[Ru(CO)_4(SiMe_3)]^z$

(ii) $[(\eta^5\text{-}C_6H_7)Fe(CO)_3]^z$

(iii) $[(\eta^6\text{-}C_6H_6)_2Ru]^z$



C_6H_7

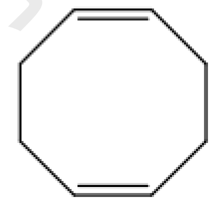
HOMEWORK

2. Propose a reasonable structure for each of the following empirical formula:

   (a) $Rh(COD)BPh_4$

   (b) $RhCl(CO)_2$

   (c) $Cp_2W(CO)_2$

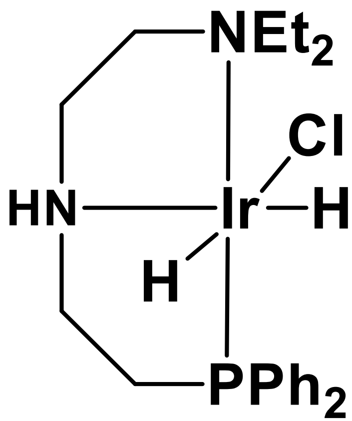


COD

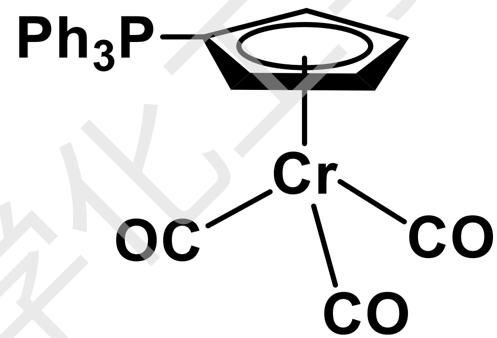
HOMEWORK

3. Which of the following complexes follows/follow the 18e rule?

(a)



(b)



(A) both of them

(B) none of them

HOMEWORK

4. Which of the following complexes is least likely to be isolated?

(a) $ReH_7(PPh_3)_2$ (b) $RuHCl(PPh_3)_3$

(c) $[Cp_2VMe_4]^+$ (d) $[WH(CH_2{=}CH_2)_4]^{3-}$

3	4	5	6	7	8	9	10	11	12
21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn
39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd
57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg
89 Ac	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn

HOMEWORK

5. On the basis of hard-soft acid-base concept, which of the following complexes would you expect to be least stable?

   (a) $Cr(CO)_6$ (b) $Pd(PMe_3)_4$

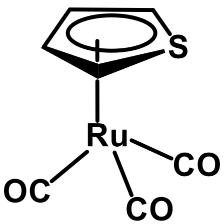
   (c) WMe_6 (d) OsO_4

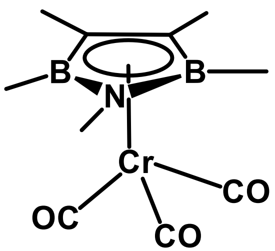
6. Which of the following complexes has the lowest ν(C-O)?

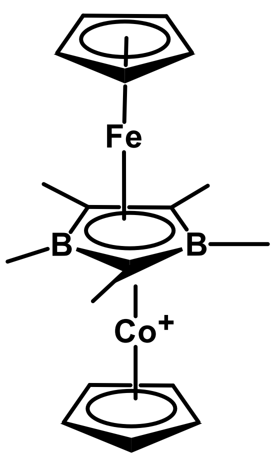
$V(CO)_6$ $Cr(CO)_6$ $Mn_2(CO)_{10}$ $Fe(CO)_5$ $CO_2(CO)_8$ $Ni(CO)_4$

HOMEWORK

7. Do the following complexes follow the 18e rule?



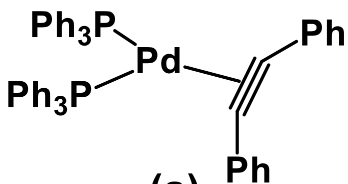




HOMEWORK

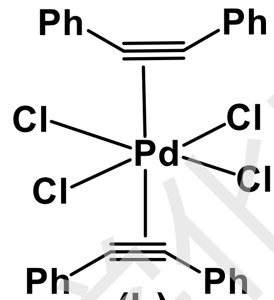
8. Which of the following complexes would you expect to be more stable?

(1)



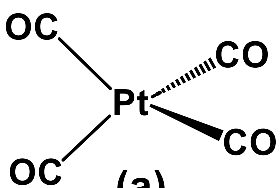
(a)

vs



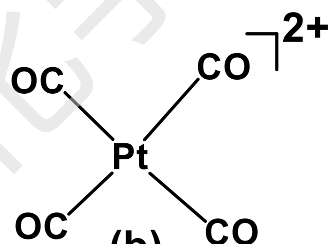
(b)

(2)



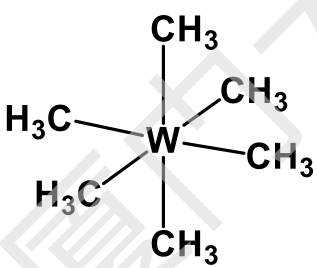
(a)

vs



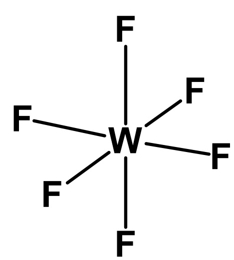
(b)

(3)



(a)

vs



(b)

HOMEWORK

Bonus Challenge:

Read the short passage on "Differences between Metals" (1.9). Why is C-H activation (breaking a C-H bond) generally easier with second and third-row transition metals (like Rh, Ir, Pd, Pt) compared to first-row metals (like Ni, Co, Fe)?